

Worksheet 2 — Reactor Systems Engineering

High School · Grades 9–12 · Engineering, Thermodynamics, Systems Thinking · ~50 min

Engineers design nuclear plants as systems of systems. In this worksheet you will analyze thermodynamic efficiency, defense-in-depth safety, and the trade-offs between competing reactor designs — the same trade-spaces practicing nuclear, mechanical, and electrical engineers navigate.

Name: _____

Date: _____

Class: _____

Part A · Thermodynamic Efficiency

The Carnot efficiency sets the upper bound for any heat engine: $\eta_{\text{Carnot}} = 1 - T_{\text{cold}} / T_{\text{hot}}$, with temperatures in Kelvin.

Reactor type	T_hot (°C)	T_cold (°C)	η_{Carnot}	Real η
PWR (Gen II)	315	30	_____	~33%
BWR (Gen II)	285	30	_____	~33%
High-Temp Gas Reactor	750	30	_____	~45%
Molten Salt Reactor	700	30	_____	~44%
Sodium Fast Reactor	550	30	_____	~40%

A1. Compute η_{Carnot} for each row. **A2.** Why is the real efficiency always far below Carnot? Identify at least three real-world loss mechanisms. **A3.** Which design is the best candidate for industrial process heat (hydrogen production, desalination, chemical synthesis) and why?

Part B · Defense in Depth

Nuclear safety relies on independent, redundant barriers. Match each safety feature with the failure it is designed to mitigate.

Safety feature	Failure it mitigates
1. Fuel pellet ceramic matrix (UO ₂)	A. Loss of off-site power (station blackout)
2. Zircaloy fuel cladding	B. Sudden reactivity insertion / power excursion
3. Reactor pressure vessel (steel, 8 in thick)	C. Release of fission products to environment
4. Containment building (reinforced concrete + steel liner)	D. Loss-of-coolant accident (LOCA)
5. Negative temperature coefficient	E. First barrier: traps fission products in ceramic
6. Emergency core cooling system (ECCS)	F. Mechanical impact, aircraft strike, external hazards
7. Passive natural-circulation cooling (Gen III+)	G. Second barrier: contains gaseous fission products

Your matches: 1__ 2__ 3__ 4__ 5__ 6__ 7__

Part C · Reactor Design Trade Study

You are a junior engineer on a team selecting a reactor for a remote Alaskan community of 5,000 people with a 20 MW peak demand. Score the three candidates on a 1–5 scale across each criterion, justify with one sentence, then recommend one design.

Criterion	Large PWR (1100 MW)	SMR (BWRX-300, 300 MW)	Microreactor (10 MW)
Capital fit to load			
Site footprint			
Refueling logistics			
Construction time			
Operator staffing			
Total (out of 25)			

C1. Which design did you recommend and why?